

6.2 Process Specification (based on Logical DFD TO-BE)

Number : 1.0 Name : Register User Account Description : UTM Administration inputs the student's registration details to create a new user profile in the User Database.	
Input Data Flow : User Registration Details from UTM Administration	
Output Data Flow : New User Profile to User Database	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ User Registration Details from UTM Administration IF Student ID is empty OR Temporary Login Credential is empty OR Email is empty THEN RETURN Registration Error to UTM Administration ELSE CHECK Student ID in User Database IF Student ID exists THEN RETURN Registration Error to UTM Administration ELSE CREATE New User Profile WRITE New User Profile to User Database RETURN Registration Successful to UTM Administration ENDIF ENDIF	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 2.0

Name : Authenticate User

Description : Receives the student's login credentials, then checks with user information in User Database and lastly returns a login access to the student.

Input Data Flow :

Login Credentials from Student

User Information from User Database

Output Data Flow :

Login Access to Student

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

READ Login Credentials from Student

FIND User Information in User Database

IF Login Credentials matched with User Information

 THEN RETURN Successful Login Access to Student

ELSE

 RETURN Failed Login Access to Student

ENDIF

Refer to: Name: _____

☒ Structured English

☐ Decision Table

☐ Decision Tree

Unresolved Issues: -

Number : 3.0

Name : Manage Bus Schedule

Description : Receives bus timetable information from UTM Administration, checks with an existing timetable record in the Bus Schedule Database, and creates a new record or updates the existing record accordingly.

Input Data Flow :

Bus Timetable Information from UTM Administration

Timetable Record from Bus Schedule Database

Output Data Flow :

New Timetable to Bus Schedule Database

Bus Timetable to UTM Administration

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

READ Bus Timetable Information from UTM Administration

RETRIEVE Timetable Record from Bus Schedule Database

SEND Bus Timetable to UTM Administration

IF Bus Timetable is found

 IF UTM Administration request to delete bus schedule

 THEN DELETE Bus Timetable from Bus Schedule Database

 DISPLAY "Bus Schedule Deleted Successfully"

 ELSE IF UTM Administration request to update bus schedule

 THEN UPDATE Bus Timetable with new information

 WRITE New Timetable to Bus Schedule Database

 ELSE IF UTM Administration request to create bus schedule

 THEN CREATE New Timetable Record

 WRITE New Timetable to Bus Schedule Database

 ELSE continue

 ENDIF

ELSE

 SEND error message to UTM Administration

ENDIF

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 4.0

Name : Retrieve Bus Schedule

Description : Retrieves bus timetable information from the Bus Schedule Database and provides it to the student after student request.

Input Data Flow :

Bus Timetable Request from Student

Timetable from Bus Schedule Database

Output Data Flow :

Latest Bus Timetable to Student

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

READ Bus Timetable Request from Student

RETRIEVE Timetable from Bus Schedule Database

IF Timetable is found

 THEN SEND Latest Bus Timetable to Student

ELSE

 SEND error message to Student

ENDIF

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 5.0

Name : Track Bus Live Location

Description : Retrieve real-time GPS coordinates from the bus transponder and provide live location and ETA updates to the student.

Input Data Flow :

Bus Selection from Student

Live GPS Coordinates from GPS Transponder

Output Data Flow :

Bus Live Location to Student

Estimated Time of Arrival to Student

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

GET Bus Selection from Student

DO WHILE Bus is Active

 GET Live GPS Coordinates from GPS Transponder

 IF GPS Coordinates are available

 THEN CALCULATE Estimated Time of Arrival

 SEND Bus Live Location to Student

 SEND Estimated Time of Arrival to Student

 ELSE

 SEND "GPS signal unavailable" to Student

 ENDIF

ENDDO

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 6.0

Name : Handle Seat Occupancy

Description : Receive real-time seat occupancy data from the seat availability sensor, update occupancy record in the database, and return the seat occupancy status to Student.

Input Data Flow :

Current Seat Occupancy Data from Seat Availability Sensor
Seat Availability from Occupancy Database

Output Data Flow :

Occupancy Record to Occupancy Database
Seat Occupancy Status to Student

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

```
DO WHILE system is operating
    GET Current Seat Occupancy Data from Seat Availability Sensor
    IF Seat Availability Sensor detects passenger boarding
        THEN INCREASE occupancy count by 1
    ELSE IF Sensor detects passenger alighting
        THEN DECREASE occupancy count by 1
    ELSE
        REMAIN current occupancy count
    ENDIF
    MOVE occupancy count to Occupancy Record
    WRITE Occupancy Record to Occupancy Database
    READ Seat Availability from Occupancy Database
    IF Seat Availability >=15
        THEN SET Seat Occupancy Status to Available
    ELSE IF Seat Availability >1 AND Seat Availability <15
        THEN SET Seat Occupancy Status to Moderate
    ELSE
        SET Seat Occupancy Status to Full
    ENDIF
    SEND Seat Occupancy Status to Student
ENDDO
```

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 7.1

Name : Parse Search Query

Description : Parse the student's location search input into structured target coordinates

Input Data Flow :

Location Search from Student

Output Data Flow :

Target Coordinates to Process 7.2

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

GET Location Search from Student

IF Location Search is NOT empty

 THEN PARSE location name into Target Coordinates (latitude, longitude)

 SEND Target Coordinates to Process 7.2

ELSE

 SEND error message to Student

ENDIF

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 7.2

Name : Query Route and Bus Data

Description : Queries the Google Maps Service using the target coordinates to retrieve route and distance data, and retrieves available campus bus routes from the Bus Schedule Database that serve the requested destination.

Input Data Flow :

Target Coordinates from Process 7.1

Map Topography Data from Google Maps Service

Calculated Distance Metrics from Google Maps Service

Bus Route Data from Bus Schedule Database

Output Data Flow :

Location Distance Request to Google Maps Service

Route Calculation Request to Google Maps Service

Search Details to Process 7.3

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

GET Target Coordinates from Process 7.1

RETRIEVE Bus Route Data from Bus Schedule Database

IF Bus Route Data covers Target Coordinates

 THEN SEND Location Distance Request to Google Maps Service based on available Bus Route Data

 SEND Route Calculation Request to Google Maps Service based on available Bus Route Data

 RECEIVE Map Topography Data from Google Maps Service

 RECEIVE Calculated Distance Metrics from Google Maps Service

 SET Search Details with Map Topography Data, Calculated Distance Metrics and Bus Route Data

 SEND Search Details to Process 7.3

ELSE

 SEND error message to Student

ENDIF

Refer to: Name: _____

☒ Structured English

☐ Decision Table

☐ Decision Tree

Unresolved Issues: -

Number : 7.3

Name : Generate Route Guidance

Description : Processes the combined route and bus data to generate complete route navigation guidance, including directions, distances, and available campus bus options, and delivers it to the student.

Input Data Flow :

Search Details from Process 7.2

Output Data Flow :

Route Navigation Guidance to Student

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

GET Search Details from Process 7.2

SET Route Navigation Guidance with directions, distance and available bus information

SEND Route Navigation Guidance to Student

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 8.1

Name : Post Ride Offer

Description : Verifies student's year of study, then receives ride details from a student driver that offers a carpool, then validates the details, and stores the offer in the carpool database.

Input Data Flow :

Ride Offer Details from Carpool Driver

Year of Study from User Database

Output Data Flow :

Trip and Driver Information to Carpool Database

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

RETRIEVE Year of Study from User Database

IF Year of Study > 1

 THEN READ Ride Offer Details from Carpool Driver

 IF destination is not empty AND Available Seats greater than 0

 WRITE Trip and Driver Information to Carpool Database

 ELSE

 SEND error message to Carpool Driver

 ENDIF

ELSE

 SEND error message to Student

ENDIF

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 8.2

Name : Handle Carpool Request

Description : Shows available rides to the student, handles their join request, and sends the passenger's contact info and a match notification to the driver.

Input Data Flow :

Ride Offer List from Carpool Database

Join-trip Request from Student

Output Data Flow :

Passenger Match Notification to Carpool Driver

Passenger Contact Information to Carpool Driver

Trip Booking & Acceptance Details to Process 8.3

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

READ Ride Offer List from Carpool Database

READ Join-trip Request from Student

IF Join-trip Request matched with Ride Offer List

 THEN SEND Passenger Match Notification to Carpool Driver

 SEND Passenger Contact Information to Carpool Driver

 CREATE Trip Booking & Acceptance Details

 SEND Trip Booking & Acceptance Details to Process 8.3

ELSE

 SEND "No matching rides found" to Student

ENDIF

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 8.3

Name : Process Matching of Carpool

Description : Processes the booking details, sends the driver's vehicle info to the student, and sends matched booking details to the next process.

Input Data Flow :

Trip Booking & Acceptance Details from Process 8.2

Output Data Flow :

Matched Driver Details to Student

Matched Booking Details to Process 8.4

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

READ Trip Booking & Acceptance Details from Process 8.2

SET Matched Driver Details with Driver Contact Number, Pickup Time and Trip Fare

SEND Matched Driver Details to Student

SEND Matched Booking Details to Process 8.4

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 8.4

Name : Finalize Trip Records

Description : Saves the final confirmed trip and passenger information into the carpool database for record keeping.

Input Data Flow :

Matched Booking Details from Process 8.3

Output Data Flow :

Passenger Information to Carpool Database

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

READ Matched Booking Details from Process 8.3

WRITE Passenger Information to Carpool Database

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 9.0

Name : Generate Feedback Report

Description : Continuously receive and save live student trip feedback into the database while the system is operational, and handles requests to retrieve stored feedback data to generate performance reports and show feedback log to UTM Administration.

Input Data Flow :

Trip Feedback Data from Student

Feedback & Complaint Data from Feedback Database

Output Data Flow :

Feedback Records to Feedback Database

Feedback Logs to UTM Administration

Monthly Performance Report to UTM Administration

Type of Process :

☐ Online ☒ Batch ☐ Manual

Subprogram/Function Name:

DO WHILE system is operating

 READ Trip Feedback Data from Student

 IF Trip Feedback Data is received

 THEN CREATE New Feedback Record

 WRITE New Feedback Record to Feedback Database

 MOVE Feedback Data to Feedback Logs

 SEND Feedback Logs to UTM Administration

 ELSE continue

 ENDIF

 IF System Date is End of Month AND Current Time is EQUAL to "23:59"

 THEN READ Feedback & Complaint Data from Feedback Database

 COMPILE historical feedback trends and complaints from that month

 CREATE Monthly Performance Report

 SEND Monthly Performance Report to UTM Administration

 ELSE continue

 ENDIF

ENDDO

Refer to: Name: _____

☒ Structured English

☐ Decision Table

☐ Decision Tree

Unresolved Issues: -